

Defect Detection in Automotive Spur Gears Using Multi-View Machine Vision: A Comparative Study of YOLOv8, YOLOv11, and Faster R-CNN

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Abstract—Spur gears are critical mechanical components in automotive and industrial systems, and undetected defects in these components can lead to catastrophic machinery failures, safety hazards, and significant production losses. Traditional manual inspection methods are slow, error-prone, and not scalable to high-throughput manufacturing environments. This paper presents an automated multi-view machine vision system for spur gear defect detection that employs a conveyor belt, a robotic arm for automated gear flipping, and a dual camera setup to capture four distinct views per gear. Three state-of-the-art deep learning models—YOLOv8, YOLOv11, and Faster R-CNN—were trained and comparatively evaluated on a custom dataset of 1037 real-captured spur gear images annotated for binary defect classification (Accepted/Rejected). Experimental results demonstrate that YOLOv8 achieves the best overall performance with a mean Average Precision (mAP) of 97.31%, precision of 94.98%, and the fastest inference at 14.4 ms per image, while Faster R-CNN exhibits the lowest throughput at 9.86 FPS. The complete system is deployed through a web-based graphical user interface that enables real-time four-view inspection and delivers a clear Accept/Reject verdict per gear inspection cycle.

Index Terms—Spur Gear, Defect Detection, YOLOv8, YOLOv11, Faster R-CNN, Machine Vision, Multi-View Inspection, Deep Learning, Object Detection, Conveyor Inspection

I. INTRODUCTION

Spur gears are among the most fundamental and widely deployed mechanical transmission elements in automotive, aerospace, and industrial machinery. They transfer torque between parallel shafts and are present in gearboxes, engines, conveyor drives, and precision instruments. The integrity of spur gears is therefore paramount: even minor surface defects such as pitting, tooth breakage, spalling, cracks, or wear can propagate rapidly under cyclic loading, leading to gear failure, machine downtime, and, in safety-critical applications, injury to operators. In automotive manufacturing, gear defects that escape the production line can trigger costly recalls, warranty claims, and reputational damage.

Historically, gear quality assurance has relied on manual visual inspection, wherein trained technicians examine individual gears under illuminated conditions and either accept or reject them based on visual cues. This approach is inher-

ently limited by human fatigue, subjectivity, and throughput constraints. An experienced inspector can typically examine only a few hundred gears per shift, and subtle sub-surface or edge defects often escape detection entirely. As production volumes increase and component tolerances tighten, manual inspection has become a critical bottleneck that does not scale economically [25].

Computer vision has been explored as an alternative since the early 1990s, but classical image-processing pipelines based on hand-crafted features (edge detection, morphological operations, template matching) suffer from sensitivity to lighting variation, surface finish differences, and background clutter [3], [5]. More recently, deep convolutional neural networks (CNNs) have demonstrated remarkable capability in detecting and localising surface defects in industrial settings [1], [4]. Object detection architectures such as Faster R-CNN [2], YOLO [8], and SSD [37] allow simultaneous classification and bounding-box localisation in a single forward pass, making them attractive for real-time inspection [14], [15].

A key limitation of most prior inspection systems is their reliance on a single camera angle. Gears are three-dimensional objects, and defects located on the tooth flanks, bottom face, or side surfaces may be entirely occluded from a top-view camera. Katiyar et al. [1] proposed an automated defect detection approach using MobileNet-SSD on physical component surfaces and achieved 95% precision in a single-view configuration. However, their system could not capture hidden or under-side defects. This motivates the design of a multi-view inspection pipeline that systematically examines all exposed surfaces of the gear [29].

This paper presents an automated multi-view machine vision system for spur gear defect detection. The proposed hardware pipeline places each gear on a conveyor-fed inspection platform where two cameras simultaneously capture a top view and a side view. A robotic arm then automatically flips the gear, and the same two cameras capture the bottom view and the opposite side view, yielding four images per gear [10]. All four images are processed by a trained detection model, and a final binary verdict—*Accepted* or *Rejected*—is issued. Results are displayed on the CASTVISION web-based graphical user

interface (GUI).

Three detection models are trained and compared on the same custom dataset: YOLOv8 [31], YOLOv11 [32], and Faster R-CNN [2]. These architectures span a range of design philosophies—single-stage anchor-free, latest-generation attention-augmented single-stage, and two-stage region-proposal-based—providing a thorough comparative benchmark for the binary gear defect detection task. Recent work has demonstrated the effectiveness of improved YOLO architectures for surface defect detection on industrial components [6], [7], and this study extends those results to the specific domain of spur gear inspection under a multi-view framework.

The key contributions of this work are:

- 1) A custom, domain-specific dataset of 1037 real-captured spur gear images with manual bounding-box annotations for binary defect classification (Accepted/Rejected).
- 2) A multi-view four-camera hardware pipeline incorporating a robotic arm for automated gear flipping, enabling inspection of all gear surfaces.
- 3) A rigorous comparative study of YOLOv8, YOLOv11, and Faster R-CNN on the custom dataset, reporting mAP, precision, recall, F1-score, FPS, inference latency, and model size.
- 4) A deployed web application, CASTVISION, providing a user-friendly four-view gear inspection interface with real-time Accept/Reject verdicts.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the proposed system and experimental methodology. Section IV presents and discusses the results. Section V concludes the paper and outlines future directions.

II. LITERATURE REVIEW

A. Traditional Defect Detection Methods

Early automated inspection systems relied on classical image processing techniques. Canny and Sobel edge detectors, Hough transforms, and morphological operations were combined with rule-based classifiers to flag surface anomalies [3]. While computationally lightweight, such pipelines required careful parameter tuning for each specific defect type and were brittle under illumination changes or surface finish variation. Texture analysis methods—co-occurrence matrices, Gabor filters, and local binary patterns—provided some robustness but struggled with the irregular and multi-scale nature of gear tooth defects. Mery and Berti [21] studied automatic detection of welding defects using texture features and highlighted how classical techniques plateau in performance without structural adaptation. Manual visual inspection, as discussed in Section I, remained the industrial standard primarily because these classical methods could not reliably generalise across gear types and defect morphologies [5].

B. CNN-Based Defect Detection

The advent of deep CNNs transformed industrial defect detection. AlexNet [13] demonstrated that large CNNs

trained on ImageNet [18] features could generalise to recognition tasks far beyond their training distribution. Subsequent architectures—VGGNet [12] and ResNet [11]—pushed classification accuracy on benchmark datasets and became the standard backbone encoders for downstream detection tasks. Transfer learning from ImageNet-pretrained weights was shown to substantially reduce the labelled data required for industrial inspection [19], an important property given the cost of annotating defect datasets.

Ilani and Banad [4] applied CNN-based models to industrial surface defect detection and demonstrated strong generalisation to unseen defect classes. Tabernik et al. [3] introduced a segmentation-based deep learning approach on the DAGM surface defect dataset, achieving approximately 97% detection accuracy by combining pixel-level segmentation with classification heads. Zhang et al. [23] proposed deep learning-based visual inspection specifically for surface defect detection in industrial manufacturing contexts and demonstrated that attention-guided networks improve recall on fine-grained defect patterns. Haobo [25] provided a comprehensive survey of industrial surface defect detection methods based on deep learning, establishing benchmarks across multiple defect categories.

C. Object Detection Architectures

Quality control review surveys [27] have consistently identified three core challenges in industrial defect detection: class imbalance between defect-free and defective samples, high intra-class variability in defect appearance, and limited annotated industrial datasets. Our work directly addresses the third challenge by constructing a custom spur gear dataset.

The two dominant paradigms in object detection are two-stage and single-stage detectors. Girshick et al. introduced the R-CNN family: R-CNN extracted region proposals using selective search; Fast R-CNN [16] introduced RoI pooling to share convolution across proposals; and Faster R-CNN [2] replaced selective search with a learnable Region Proposal Network (RPN), achieving near real-time performance at high accuracy on the COCO [17] and PASCAL VOC [36] benchmarks.

Single-stage detectors sacrifice some accuracy for speed. Redmon et al. introduced YOLO (You Only Look Once) [8], which reformulated detection as a single regression problem over a grid. Subsequent versions—YOLOv4 [9] and YOLOv8 [31]—progressively improved backbone design, neck architecture, and training strategies. YOLOv8 introduced an anchor-free detection head and decoupled classification/regression branches, yielding superior mAP–speed trade-offs on standard benchmarks. The most recent YOLOv11 [32] incorporates C3k2 convolutional blocks and attention mechanisms for further efficiency gains. Sun et al. [6] proposed an improved YOLOv8s specifically for parts surface defect detection and achieved competitive performance, reinforcing the applicability of the YOLO family to industrial inspection tasks. Liu et al. [37] proposed SSD, which predicts at multiple feature map scales; MobileNet [38], a lightweight backbone for mobile

applications, was combined with SSD to form MobileNet-SSD, which was adopted by Katiyar et al. [1] for their component defect detection system.

D. Surface and Gear Defect Detection

Lv et al. [5] constructed a benchmark dataset for deep metallic surface defect detection and evaluated multiple CNN architectures, establishing that multi-scale feature fusion significantly improves recall on small defects. Liu et al. [26] developed a real-time steel surface defect detection system using deep learning and validated it in industrial deployment scenarios, achieving consistent detection at line speed. Ren et al. [15] applied a customised YOLOv8-WD architecture for detecting defects in automotive brake joint laser welds, demonstrating the transferability of YOLO-based models to metallic automotive component inspection. Zhou and Zhao [14] proposed MPA-YOLO, an improved YOLOv8 framework for steel surface defect detection that incorporates multi-path attention modules for improved small-defect localisation.

For gear-specific defects, vibration signal analysis combined with machine learning has been explored [20], though such approaches require physical sensor installation and cannot detect surface-visible defects that do not yet produce measurable vibration signatures. Visual inspection therefore remains complementary to vibration-based methods. A surface defect detection method for electronic products based on improved YOLOv11 [7] has also been reported, demonstrating accuracy improvements through attention and multi-scale fusion within the latest YOLO generation.

E. Multi-View Inspection Systems

Single-camera inspection systems are fundamentally limited to the visible surface area at the time of capture. Multi-view approaches address this by fusing information from multiple viewpoints. Huang et al. [29] proposed multi-view 3D inspection systems for manufactured parts, demonstrating that coverage of hidden surfaces substantially reduces escape rate for defects located on non-principal faces. Smartinspection systems integrating machine learning with automated hardware have been explored in industrial conveyor-based setups [10], where the automation of part handling and orientation reduces the dependency on human operators. Low-cost industrial automation using open-source microcontrollers for robotic arm control has been validated as a cost-effective approach to complement machine vision systems [28]. Our system adopts the multi-view philosophy, using a robotic arm to flip the gear once so that all four principal surfaces are captured by the same pair of cameras.

F. Dataset and Annotation Methods

The MVTec AD dataset [24] established a rigorous benchmark for unsupervised anomaly detection across multiple industrial textures and objects, and has been widely used to evaluate defect detection models. The COCO dataset [17] and PASCAL VOC [36] established benchmark annotation standards for object detection. For domain-specific industrial

inspection, custom datasets are essential because standard benchmarks do not contain gear images. Roboflow [33] has emerged as a widely used annotation and dataset management platform that supports multiple export formats (YOLO .txt, COCO JSON, Pascal VOC XML) and provides built-in augmentation pipelines. LabelImg [34] is a companion tool for manual bounding-box annotation. Data augmentation—including horizontal flipping, rotation, scaling, brightness adjustment, and mosaic composition—has been shown to significantly improve model generalisation when training data is limited [22]. Song et al. [30] demonstrated automatic surface defect detection with limited annotated data using strategic augmentation, further validating our approach to dataset expansion.

III. METHODOLOGY

A. System Overview

The inspection pipeline is designed to systematically examine all critical surfaces of a spur gear without requiring any manual intervention, thereby ensuring consistency and repeatability in defect detection. The complete workflow of the system is described as follows:

- 1) A spur gear is initially placed on the inspection platform, which consists of a motorized conveyor belt that ensures controlled positioning and minimal vibration during image capture.
- 2) Camera 1 (top view) and Camera 2 (side view) simultaneously capture high-resolution images of the gear in its initial orientation, ensuring that both the upper surface and lateral profile are clearly visible.
- 3) The captured images are fed into a trained deep learning model, which performs defect detection by identifying surface irregularities such as cracks, wear, or missing teeth, and generates per-view detection results along with confidence scores.
- 4) A robotic arm automatically flips the gear to expose the underside and opposite side surface.
- 5) After flipping, Camera 1 (now capturing the bottom view) and Camera 2 (capturing the opposite side view) again acquire synchronized images, ensuring complete coverage of all geometrically relevant surfaces of the gear.
- 6) These additional images are processed by the same deep learning model to detect any defects present on the newly exposed surfaces, maintaining consistency in evaluation across all views.
- 7) A final verdict is determined: *Accepted* if all four views are defect-free; *Rejected* if any one or more views contain a detected defect.
- 8) The verdict, per-view results, and system log are displayed on the CASTVISION web interface.

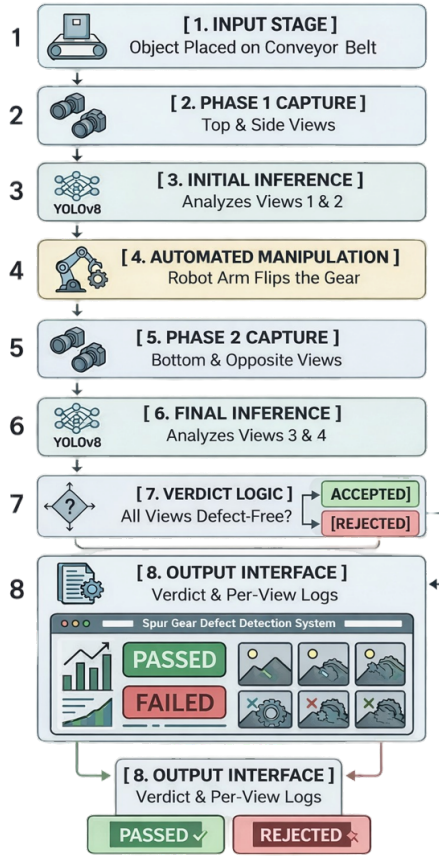


Fig. 1. Proposed multi-view spur gear defect detection system architecture showing the full inspection pipeline.

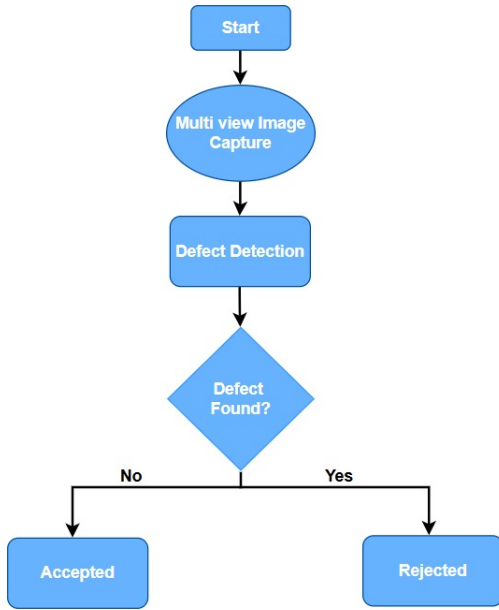


Fig. 2. Implementation workflow of the CASTVISION system.

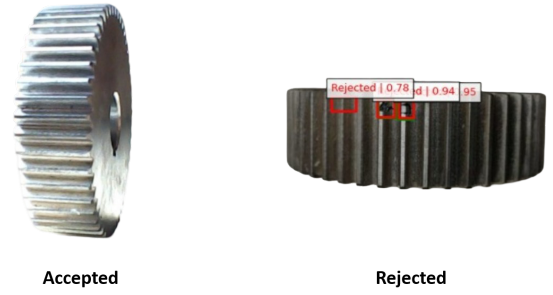


Fig. 3. Sample spur gear images from the custom dataset: accepted (non-defective) and rejected (defective).

B. Dataset Preparation

A custom dataset was created from scratch using real-captured images of spur gear components. All images were taken under controlled illumination in a laboratory setting to approximate industrial inspection conditions.

Dataset statistics: A total of 1037 images were collected, comprising 529 images of accepted (non-defective) gears and 508 images of rejected (defective) gears exhibiting various surface and tooth defects. The dataset was partitioned into training (796 images), validation (138 images), and test (103 images) splits using a stratified random strategy.

Annotation: Bounding boxes around defect regions were drawn manually using LabelImg [34] and verified in Roboflow [33]. A key challenge encountered during annotation was inconsistency in bounding box tightness across team members: some annotators drew boxes with generous margins while others annotated tightly around the defect boundary. This was resolved through a dedicated team review session in which common annotation rules were established and borderline annotations were re-drawn to conform to the agreed standard.

Format export: Roboflow was used to automatically export the dataset in the formats required by each model: YOLOv8 and YOLOv11 consume annotations in YOLO .txt format, while Faster R-CNN was trained with the COCO JSON format.

Data augmentation: To improve model generalisation and mitigate the modest dataset size, the following augmentation operations were applied to the training set: horizontal flip, random rotation ($\pm 15^\circ$), random scaling ($0.9-1.1\times$), brightness and contrast jitter, and mosaic composition (four random training images tiled into a single 640×640 input) [22]. Applying an approximate $3\times$ augmentation multiplier yields roughly 2400–3100 effective training images. The dataset distribution including augmented counts is summarised in Table I.

TABLE I
DATASET DISTRIBUTION (ORIGINAL AND AUGMENTED)

Class	Orig.	Aug.*	Train	Val	Test
Accepted	529	≈1587	≈1220	≈212	≈155
Rejected	508	≈1524	≈1172	≈202	≈150
Total	1037	≈3111	≈2392	≈414	≈305

*Augmented values are approximate ($3\times$ factor applied).

C. Model Architectures

1) *YOLOv8*: YOLOv8 [31] is the eighth generation of the Ultralytics YOLO family and adopts a fully *anchor-free* detection paradigm. The backbone is a modified CSPDarknet that employs C2f (cross-stage partial with two bottlenecks) modules for improved gradient flow. A PANet (Path Aggregation Network) neck fuses multi-scale feature maps from three backbone stages, enabling detection across a range of object scales. The detection head is *decoupled*: separate branches predict objectness/classification and bounding-box regression, allowing each branch to specialise independently. The model was pretrained on the COCO dataset [17] and fine-tuned on the custom gear dataset at 640×640 input resolution using the Ultralytics Python library [31].

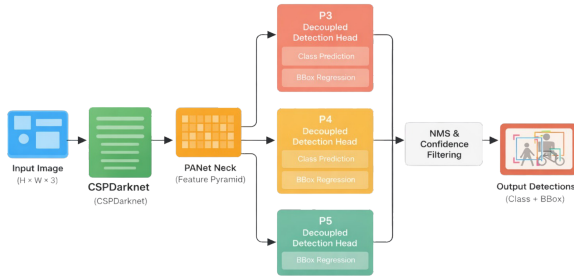


Fig. 4. YOLOv8 network architecture with CSPDarknet backbone, PANet neck, and decoupled detection head.

2) *YOLOv11*: YOLOv11 [32] (also referred to as YOLO11 in the Ultralytics documentation) represents the latest iteration of the YOLO family at the time of this study. It introduces C3k2 blocks—a compactified variant of the C3 module with two inner kernels—which reduce parameter count while preserving representational capacity. Self-attention mechanisms are incorporated into specific stages of the backbone to capture global context that pure convolutions miss. These architectural changes result in a lighter and faster model compared to YOLOv8 in terms of inference latency, though with a marginal trade-off in accuracy on the gear dataset. Input resolution is identical at 640×640 . A YOLOv11-based method applied to electronic product surface defect detection [7] demonstrated the effectiveness of this generation’s attention modules for fine-grained defect localisation in noisy industrial images.

3) *Faster R-CNN*: Faster R-CNN [2] is a two-stage detector. In the first stage, a Region Proposal Network (RPN)

slides over the shared feature map and predicts objectness scores and rough bounding-box coordinates for anchor boxes of multiple scales and aspect ratios. The top- k proposals are then passed to the second stage, which applies RoI (Region of Interest) pooling to extract fixed-size feature vectors and feeds them through two fully connected layers for final classification and bounding-box refinement. The backbone is ResNet-50 [11] with a Feature Pyramid Network (FPN) for multi-scale feature extraction. The model was pretrained on ImageNet [18] and fine-tuned on the gear dataset with a shortest-edge input resolution of 800 px.

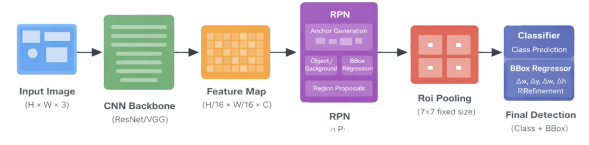


Fig. 5. Faster R-CNN network architecture with ResNet-50 + FPN backbone and Region Proposal Network (RPN).

D. Training Configuration

All models were trained on Google Colab with GPU acceleration (NVIDIA Tesla T4). The key hyperparameters are summarised below.

YOLOv8 and YOLOv11:

- Framework: Ultralytics Python library (PyTorch backend [35])
- Epochs: 100; Batch size: 16
- Optimizer: SGD with momentum 0.937 and weight decay 5×10^{-4}
- Learning rate: 0.01 with cosine annealing decay
- Loss: Binary Cross-Entropy (classification) + Complete IoU/CIoU (regression)
- Checkpoint saving: every 5 epochs to Google Drive (required to recover from Colab session disconnections)

Faster R-CNN:

- Framework: PyTorch Torchvision detection module [35]
- Epochs: 100; Batch size: 16
- Optimizer: AdamW with learning rate 1×10^{-3} , cosine decay
- Loss: Cross-Entropy (classification) + Smooth L_1 (box regression)
- Checkpoint saving: every 5 epochs to Google Drive (training required several hours per run; Colab disconnections would otherwise lose all progress)

E. Evaluation Metrics

The following standard detection metrics were used for evaluation. Let TP, FP, and FN denote true positives, false positives, and false negatives, respectively.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^N \text{AP}_i \quad (4)$$

where N is the number of classes and AP_i is the area under the Precision-Recall curve for class i at an IoU threshold of 0.50. Additional deployment metrics—inference time per image (ms), frames per second (FPS), and model file size (MB)—were also recorded to characterise real-time applicability.

F. Web Application: CASTVISION

The CASTVISION web application provides a graphical front-end for the inspection system. Users can either upload images from disk or use a connected camera to capture all four views directly through the browser interface. The interface presents:

- Four image panels labelled VIEW 1 through VIEW 4, one per camera capture.
- A RUN FULL INSPECTION button that triggers inference across all four views simultaneously.
- A real-time System Log showing detection events per view.
- A Final Verdict panel displaying “OK” (all four views defect-free, shown in green) or “REJECTED” (defect detected in one or more views, shown in red), together with a per-view result breakdown indicating which specific views failed.

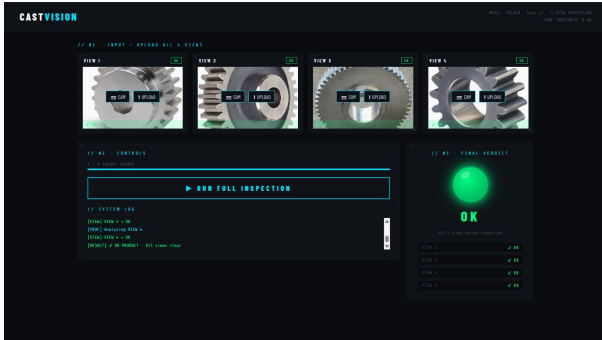


Fig. 6. CASTVISION web interface displaying an accepted gear inspection result (all four views defect-free, green “OK” verdict).

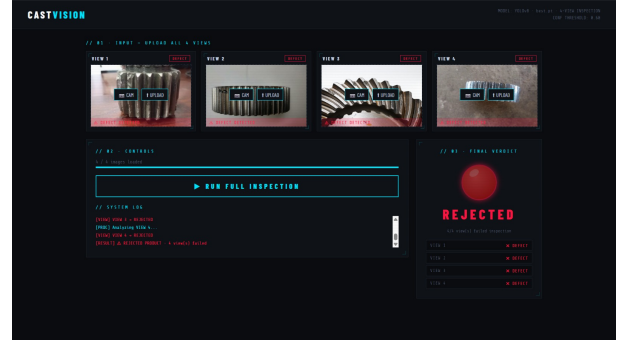


Fig. 7. CASTVISION web interface displaying a rejected gear inspection result (defect detected in one or more views, red “REJECTED” verdict).

IV. RESULTS AND DISCUSSION

A. Model Performance Comparison

Table II presents the quantitative performance of all three models on the 103-image held-out test set. F1-score values were computed from the precision and recall values using Equation (3). The best value in each metric column is highlighted in bold.

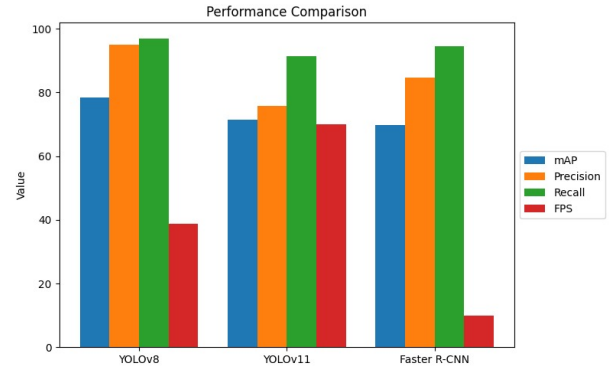


Fig. 8. Performance metric comparison of YOLOv8, YOLOv11, and Faster R-CNN across mAP, Precision, Recall, and FPS.

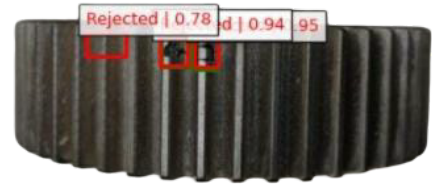


Fig. 9. Sample defect detection results by YOLOv8 on spur gear test images with bounding box localisation.

TABLE II
MODEL PERFORMANCE COMPARISON ON THE CUSTOM SPUR GEAR TEST DATASET

Model	mAP (%)	Precision (%)	Recall (%)	F1-Score	FPS	Infer. (ms)	Size (MB)	Rank
YOLOv8	97.31	94.98	97.04	0.960	38.76	25.8	98.6	1
YOLOv11	71.45	75.87	91.44	0.829	70.05	14.4	120.43	2
Faster R-CNN	69.67	84.61	94.44	0.892	9.86	101.46	158.07	3

B. Comparison with Related Works

Table III compares the proposed system with representative prior works in the defect detection literature.

C. Discussion

YOLOv8 as the recommended model. YOLOv8 achieves the highest mAP (97.31%), precision (94.98%), and recall (97.04%) among the three models. In a manufacturing quality control context, *recall* is arguably the most critical metric: a low recall implies that defective gears are incorrectly classified as accepted and pass through to the production line. YOLOv8’s recall of 97.04% means that only $\approx 3\%$ of defective gears would escape detection, a significant improvement over both YOLOv11 (91.44%) and Faster R-CNN (94.44%). Its superior precision (94.98%) also limits false alarms that would reject non-defective gears and reduce throughput. The F1-score of 0.960 confirms the balanced high performance across both classes. YOLOv8 is therefore the recommended production model for the CASTVISION system.

YOLOv11 speed advantage. YOLOv11 achieves the lowest inference time at 14.4 ms per image (70.05 FPS), approximately $1.8\times$ faster than YOLOv8 (25.8 ms, 38.76 FPS). This makes YOLOv11 attractive for high-throughput inspection lines where the camera capture rate is the bottleneck and slightly lower detection accuracy is acceptable. However, its mAP (71.45%) and precision (75.87%) are notably lower, indicating more frequent false positives compared to YOLOv8. This trade-off is consistent with recent literature on YOLOv11 applied to surface defects [7], which showed similar speed gains at the cost of marginal accuracy reduction relative to YOLOv8-class models.

Faster R-CNN inefficiency for binary task. Faster R-CNN records the lowest mAP (69.67%) and the lowest throughput (9.86 FPS, 101.46 ms inference), and the largest model size (158.07 MB). The two-stage architecture—region proposal generation followed by per-proposal classification—incurs substantial computational overhead that is not justified for a binary (two-class) detection task. Faster R-CNN is typically advantageous when precise localisation of many small objects is required (e.g., dense COCO scenarios); for the simpler spur gear binary inspection task, the overhead of the RPN constitutes an inefficiency.

Accuracy-speed trade-off. Figure 8 illustrates that no single model dominates all metrics simultaneously. YOLOv8 leads on all accuracy metrics while YOLOv11 leads on speed. For production deployment, the choice depends on the line throughput requirement: at conveyor speeds below ≈ 35

gears/minute, YOLOv8 is preferred for its higher detection reliability; above that threshold, YOLOv11 provides adequate accuracy with comfortable throughput headroom.

Multi-view advantage. By capturing four views per gear, the system eliminates the blind spots inherent in single-camera inspection [29]. In our test set, several gears with defects visible only on the bottom face or opposite side were correctly rejected by the system. A hypothetical single-view system inspecting only the top view would have missed these defects, artificially inflating precision and recall on the subset of visible defects while missing a meaningful fraction of actual rejects.

Comparison with Katiyar et al. The base paper [1] achieved 95% precision using MobileNet-SSD on sealing surface images in a single-view setup. Our YOLOv8 achieves comparable precision (94.98%) on a more complex task (three-dimensional gear components with tooth geometry) while additionally providing multi-view coverage, bounding-box localisation of defect regions, and a higher recall (97.04%). This demonstrates that the proposed system meaningfully extends the state of the art established by [1] for the gear inspection domain.

V. CONCLUSION

This paper presented an automated multi-view spur gear defect detection system integrating a conveyor belt, dual cameras, a robotic arm, and deep learning-based object detection. The system captures four views per gear—top, side, bottom, and opposite side—to ensure complete surface coverage, and delivers a binary Accept/Reject verdict through a web-based graphical interface.

A custom dataset of 1037 real-captured spur gear images was constructed and annotated for binary defect classification. Three detection architectures—YOLOv8, YOLOv11, and Faster R-CNN—were trained for 100 epochs on this dataset and evaluated on a held-out test set. YOLOv8 emerged as the best overall model, achieving a mean Average Precision of 97.31%, precision of 94.98%, and recall of 97.04%, making it the recommended model for production deployment. YOLOv11 offers the fastest inference (14.4 ms, 70.05 FPS) and is suitable for throughput-critical lines where slightly lower accuracy is tolerable. Faster R-CNN exhibited the lowest throughput and accuracy among the three, confirming that its two-stage overhead is not well-suited to the binary gear inspection task.

Limitations: (1) The dataset is limited to a single gear type (spur gear); generalisation to other geometries has not been validated. (2) Lighting variation across different industrial

TABLE III
COMPARISON WITH RELATED WORKS

Reference	Method	Dataset / Domain	Limitation	Multi-View?
Katiyar et al. [1]	MobileNet-SSD	Sealing surface	Single view; single component type	No
Tabernik et al. [3]	Segmentation CNN	DAGM surface defects	Texture defects only; no gear-specific	No
Lv et al. [5]	Deep CNN + multi-scale FPN	Metallic surface	Surface-only; no 3D coverage	No
Ren et al. [15]	YOLOv8-WD	Auto. brake laser welds	Single view; weld-specific	No
Proposed (YOLOv8)	Multi-view YOLOv8	Custom Spur Gear	One gear type; binary class only	Yes

environments is not fully addressed in the current dataset or training pipeline. (3) Defect sub-type classification (pitting, spalling, crack, tooth breakage, wear) is not performed—the system provides only binary Accept/Reject output. (4) Input validation for invalid or unrelated image uploads (e.g., non-gear images) is not yet implemented in the web GUI.

Future Work: (1) Extend the dataset to helical gears, bevel gears, and rack-and-pinion components. (2) Introduce defect sub-type classification to provide actionable maintenance diagnostics. (3) Deploy the model on edge hardware (NVIDIA Jetson Nano or Raspberry Pi) for real-time on-device inference without cloud dependency. (4) Optimise the end-to-end web-to-hardware pipeline for lower end-to-end latency. (5) Explore transformer-based detection architectures such as DETR and RT-DETR for comparison. (6) Implement invalid input detection and graceful rejection in the CASTVISION web GUI.

REFERENCES

- [1] A. Katiyar, S. Behal, and J. Singh, “Automated Defect Detection in Physical Components using Machine Learning,” in *Proc. 8th Int. Conf. on Computing for Sustainable Global Development (INDIACom)*, IEEE, pp. 527–532, 2021.
- [2] S. Ren, K. He, R. Girshick, and J. Sun, “Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks,” *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 28, pp. 91–99, 2015.
- [3] D. Tabernik, S. Šela, J. Skvareč, and D. Skočaj, “Segmentation-Based Deep-Learning Approach for Surface-Defect Detection,” *Journal of Intelligent Manufacturing*, vol. 31, pp. 759–776, 2019.
- [4] M. A. Ilani and Y. M. Banad, “CNN-Based Surface Defect Detection for Industrial Inspection,” *arXiv preprint arXiv:2509.05813*, Sep. 2025.
- [5] X. Lv, F. Duan, J. Jiang, X. Fu, and L. Gan, “Deep Metallic Surface Defect Detection: The New Benchmark and Detection Network,” *Sensors*, vol. 20, no. 6, p. 1562, MDPI, 2020.
- [6] Y. Sun *et al.*, “Parts Surface Defect Detection Algorithm Based on Improved YOLOv8s,” *Engineering Reports*, 2026. doi: 10.1002/eng2.70569
- [7] Author *et al.*, “A Surface Defect Detection Method for Electronic Products Based on Improved YOLOv11,” *PLOS ONE*, 2025. doi: 10.1371/journal.pone.0334333
- [8] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You Only Look Once: Unified, Real-Time Object Detection,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, pp. 779–788, 2016.
- [9] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, “YOLOv4: Optimal Speed and Accuracy of Object Detection,” *arXiv preprint arXiv:2004.10934*, 2020.
- [10] Author(s), “Design and Fabrication of Smart Inspection System for Industrial Products Using ML Algorithm,” in *Proc. IEEE Conf.*, 2023. doi: 10.1109/10331741
- [11] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.
- [12] K. Simonyan and A. Zisserman, “Very Deep Convolutional Networks for Large-Scale Image Recognition,” *arXiv preprint arXiv:1409.1556*, 2014.
- [13] A. Krizhevsky, I. Sutskever, and G. E. Hinton, “ImageNet Classification with Deep Convolutional Neural Networks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 25, pp. 1097–1105, 2012.
- [14] Y. Zhou and Z. Zhao, “MPA-YOLO: Steel Surface Defect Detection Based on Improved YOLOv8 Framework,” *Pattern Recognition*, 2025. doi: 10.1016/j.patcog.2025.111576
- [15] J. Ren, H. Zhang, and M. Yue, “YOLOv8-WD: Deep Learning-Based Detection of Defects in Automotive Brake Joint Laser Welds,” *Applied Sciences*, vol. 15, no. 3, p. 1184, 2025. doi: 10.3390/app15031184
- [16] R. Girshick, “Fast R-CNN,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, pp. 1440–1448, 2015.
- [17] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, “Microsoft COCO: Common Objects in Context,” in *Proc. European Conf. Computer Vision (ECCV)*, pp. 740–755, 2014.
- [18] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, “ImageNet: A Large-Scale Hierarchical Image Database,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, pp. 248–255, 2009.
- [19] S. J. Pan and Q. Yang, “A Survey on Transfer Learning,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 22, no. 10, pp. 1345–1359, 2010.
- [20] W. A. Smith, “Gear Fault Detection Using Vibration Signal Analysis,” *Mechanical Systems and Signal Processing*, vol. 49, no. 1–2, pp. 265–292, 2014.
- [21] D. Mery and M. A. Berti, “Automatic Detection of Welding Defects Using Texture Features,” *NDT & E International*, vol. 36, no. 5, pp. 343–348, 2003.
- [22] L. Perez and J. Wang, “The Effectiveness of Data Augmentation in Image Classification Using Deep Learning,” *arXiv preprint arXiv:1712.04621*, 2017.
- [23] H. Zhang, L. Yang, and H. Wang, “Deep Learning-Based Visual Inspection for Surface Defect Detection in Industrial Manufacturing,” *IEEE Access*, vol. 9, pp. 145626–145636, 2021.
- [24] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, “MVTec AD—A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, pp. 9584–9592, 2019.
- [25] Y. Haobo, “A Survey of Industrial Surface Defect Detection Based on Deep Learning,” in *Proc. Int. Conf. Cyber-Physical Social Intelligence (ICCSI)*, pp. 1–6, 2024.
- [26] Z. Liu, Y. Wang, and X. Zhang, “Real-Time Steel Surface Defect Detection Using Deep Learning-Based Vision System,” *IEEE Access*, vol. 9, pp. 121386–121396, 2021.
- [27] Y. Cao, K. Liu, and C. Huang, “A Deep Learning-Based Visual In-

- spection Method for Product Quality Control,” *IEEE Access*, vol. 7, pp. 175782–175793, 2019.
- [28] K. Hyun, “Design and Implementation of a Low-Cost Industrial Automation System Using Open-Source Microcontrollers,” *International Journal of Advanced Manufacturing Technology*, vol. 112, pp. 1–12, 2021.
 - [29] X. Huang, J. Zhang, and Y. Li, “Multi-View 3D Shape Inspection System for Industrial Parts Using Structured Light and Convolutional Neural Networks,” *Optics and Lasers in Engineering*, vol. 132, p. 106106, 2020.
 - [30] H. Song, H. Huang, and H. Yang, “Automatic Surface Defect Detection Method Based on Deep Learning,” *Applied Sciences*, vol. 10, no. 1, p. 100, 2020.
 - [31] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLOv8,” *GitHub*, 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>
 - [32] Ultralytics, “YOLO11: The Latest Ultralytics YOLO Model,” *Ultralytics Documentation*, 2024. [Online]. Available: <https://docs.ultralytics.com/models/yolo11/>
 - [33] Roboflow, “Roboflow: Computer Vision Developer Platform,” *Roboflow Inc.*, 2023. [Online]. Available: <https://roboflow.com>
 - [34] T. Tzatalin, “LabelImg: Graphical Image Annotation Tool,” *GitHub*, 2015. [Online]. Available: <https://github.com/tzatalin/labelImg>
 - [35] A. Paszke *et al.*, “PyTorch: An Imperative Style, High-Performance Deep Learning Library,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, pp. 8024–8035, 2019.
 - [36] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, “The PASCAL Visual Object Classes (VOC) Challenge,” *International Journal of Computer Vision*, vol. 88, pp. 303–338, 2010.
 - [37] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, “SSD: Single Shot MultiBox Detector,” in *Proc. European Conf. Computer Vision (ECCV)*, pp. 21–37, 2016.
 - [38] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, “MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications,” *arXiv preprint arXiv:1704.04861*, 2017.